

■ 1.8.2 Specifying Options

When *Mathematica* plots a graph, it has to make many choices. It has to work out what the scales should be, where the function should be sampled, and so on. Most of the time, *Mathematica* will probably make pretty good choices. However, if you want to get the very best possible pictures for your particular purposes, you may have to help *Mathematica* in making some of its choices.

There is a general mechanism for specifying “options” in *Mathematica* functions. Each option has a definite name. As the last argument to a function like `Plot`, you can include a sequence of rules of the form *name*->*value*, which give the values of various options. Any option for which you do not give an explicit replacement is taken to have its “default” value.

<i>option name</i>	<i>default value</i>	
PlotRange	Automatic	the range of coordinates to include in the plot: you can give $\{ymin, ymax\}$ or $\{\{xmin, xmax\}, \{ymin, ymax\}\}$; All tells <i>Mathematica</i> to include all points
PlotLabel	None	an expression to be printed as a label for the plot
Framed	False	whether to draw a frame around the plot
AspectRatio	1/GoldenRatio	the height-to-width ratio for the plot; Automatic sets it from the absolute x and y coordinates
Axes	Automatic	what axes to include: None gives no axes; $\{x, y\}$ specifies an axis origin
AxesLabel	None	labels to be put on the axes: $ylabel$ specifies a label for the y axis, $\{xlabel, ylabel\}$ for both axes
Ticks	Automatic	$\{xtick, ytick\}$ gives positions of x and y tick marks: each can be None for no tick marks, $\{t_1, t_2, \dots\}$ for explicit positions
PlotColor	True	whether to generate color graphics
DisplayFunction	\$DisplayFunction	how to display graphics; Identity causes no display
PlotStyle	Automatic	a list of lists of graphics primitives to use for each curve (see Section 1.8.13)
PlotPoints	25	the minimum number of points at which to sample the function
MaxBend	10.	the maximum kink angle between successive segments of a curve
PlotDivision	20.	the maximum factor by which to subdivide in sampling the function

Some of the options for Plot. The first set of options can also be used in Show.

A function like Plot has many options that you can set. Usually you will have to use at most a few of these at a time. If you want to optimize a particular plot,

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; <http://www.wolfram.com/>; 1-800-441-6284.

© 1988 Wolfram Research, Inc. Permission is hereby granted for web users to make one paper copy of this page for their personal use. Further reproduction, or any copying of machine-readable files (including this one) to any server computer, is strictly prohibited.

you will probably do best to “experiment”, trying a sequence of different settings for various options.

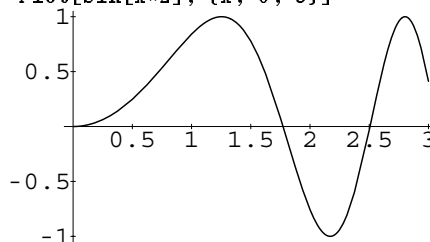
Each time you produce a plot, you can specify options for it. The next section will also discuss how you can change the options, even once you have produced a plot.

Automatic	use internal algorithms
None	do not include this
All	include everything
True	do this
False	do not do this

Some common settings for various options.

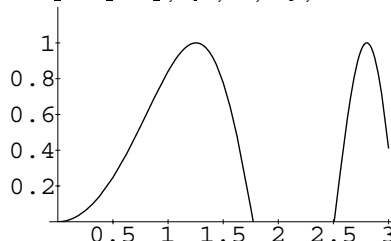
A plot with all options having their default values.

```
In[1]:= Plot[Sin[x^2], {x, 0, 3}]
```

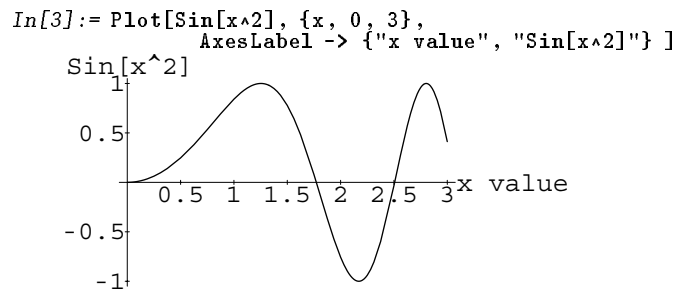


The replacement for the option `PlotRange` gives explicit y limits for the graph. With the y limits specified, the bottom of the curve is cut off. Remember to use capital letters for *each* word when you give options like `PlotRange` that have multi-word names.

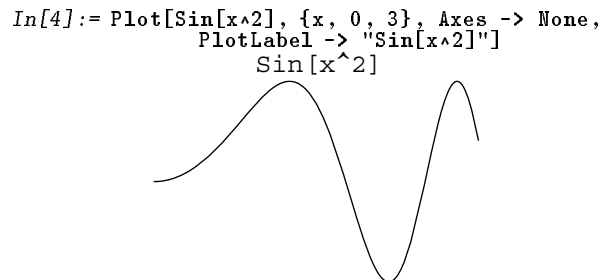
```
In[2]:= Plot[Sin[x^2], {x, 0, 3}, PlotRange -> {0, 1.2}]
```



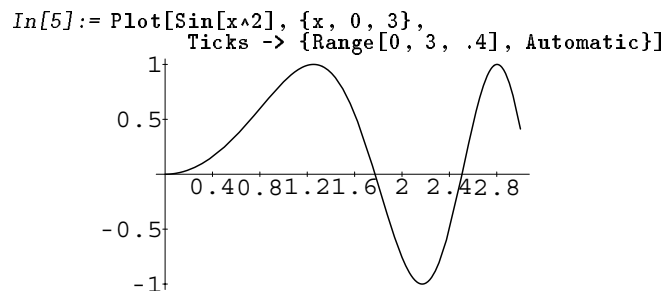
This specifies labels for the x and y axes. The expressions you give as labels are printed just as they would be if they appeared as *Mathematica* output. You can give any piece of text by putting it inside a pair of double quotes.



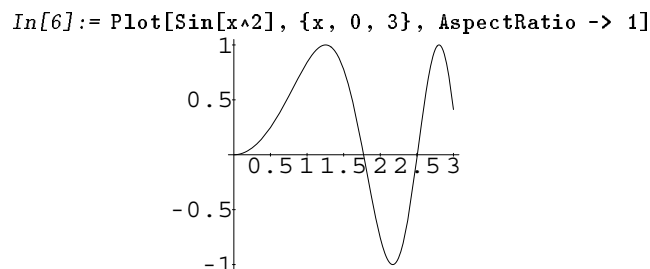
You can give several options at the same time, in any order. `Axes -> None` suppresses the drawing of axes; setting `PlotLabel` sets up a label for the graph.



This puts tick marks at intervals of 0.4 on the x axis, and leaves the positioning of tick marks on the y axis up to *Mathematica*. Remember that `Range[0, 3, .4]` produces a list of numbers running from 0 to 3 in steps of 0.4.



Setting the `AspectRatio` option changes the whole shape of your plot. `AspectRatio` gives the ratio of width to height. Its default value is the inverse of the Golden Ratio – supposedly the most pleasing shape for a rectangle.



When *Mathematica* makes a plot, it tries to set the x and y scales to include only the “interesting” parts of the plot. If your function increases very rapidly, or has singularities, the parts where it gets too large will be cut off. By specifying the option `PlotRange`, you can control exactly what ranges of x and y coordinates are

included in your plot.

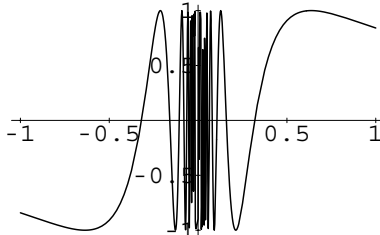
<code>Automatic</code>	show at least a large fraction of the points, including the “interesting” region (the default setting)
<code>All</code>	show all points
<code>{ymin, ymax}</code>	show a specific range of y values
<code>{xrange, yrange}</code>	show the specified ranges of x and y values

Settings for the option `PlotRange`.

Mathematica always tries to plot functions as smooth curves. As a result, in places where your function wiggles a lot, *Mathematica* will use more points. In general, *Mathematica* tries to *adapt* its sampling of your function to the form of the function. There is, however, a limit, which you can set, to how finely *Mathematica* will ever sample a function.

The function $\sin(\frac{1}{x})$ wiggles infinitely often when $x \simeq 0$. *Mathematica* tries to sample more points in the region where the function wiggles a lot, but it can never sample the infinite number that you would need to reproduce the function exactly. As a result, there are slight glitches in the plot.

```
In[7]:= Plot[Sin[1/x], {x, -1, 1}]
```



It is important to realize that since *Mathematica* can only sample your function at a limited number of points, it can always miss features of the function. By increasing `PlotPoints`, you can make *Mathematica* sample your function at a larger number of points. Of course, the larger you set `PlotPoints` to be, the longer it will take *Mathematica* to plot any function, even a smooth one.